## Chapter 19, Problem 58P

### Problem

Design a problem to help other students better understand how to develop the *y* parameters and transmission parameters, given equations in terms of the hybrid parameters.

### Step-by-step solution

**Step 1** of 2

Design a problem to help other students better understand how to develop the y parameters and transmission parameters, given equations in terms of the hybrid parameters.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

**Problem**

A two-port is described by

$\mathbf{V}_1 = \mathbf{I}_1 + 2\mathbf{V}_2, \quad \mathbf{I}_2 = -2\mathbf{I}_1 + 0.4\mathbf{V}_2$

Find: (a) the *y* parameters, (b) the transmission parameters.

**Solution**

The given set of equations is for the h parameters.

$$[\mathbf{h}] = \begin{bmatrix} 1\,\Omega & 2 \\ -2 & 0.4\text{ S} \end{bmatrix} \qquad \Delta_h = (1)(0.4) - (2)(-2) = 4.4$$

(a) $$[\mathbf{y}] = \begin{bmatrix} \dfrac{1}{\mathbf{h}_{11}} & \dfrac{-\mathbf{h}_{12}}{\mathbf{h}_{11}} \\ \dfrac{\mathbf{h}_{21}}{\mathbf{h}_{11}} & \dfrac{\Delta_h}{\mathbf{h}_{11}} \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 4.4 \end{bmatrix}\text{S}$$

**Step 2** of 2

(b) $$[\mathbf{T}] = \begin{bmatrix} \dfrac{-\Delta_h}{\mathbf{h}_{21}} & \dfrac{-\mathbf{h}_{11}}{\mathbf{h}_{21}} \\ \dfrac{-\mathbf{h}_{22}}{\mathbf{h}_{21}} & \dfrac{-1}{\mathbf{h}_{21}} \end{bmatrix} = \begin{bmatrix} 2.2 & 0.5\,\Omega \\ 0.2\text{ S} & 0.5 \end{bmatrix}$$